

Ex04. Introduction to Signals and systems

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1.1. Plot 3 pairs of Fourier Transform

This report illustrates three pairs of Fourier Transform pairs corresponding to $a = 0.1, 0.5, \text{ and } 1.0$. For numerical computation, the time domain t and frequency domain ω are sampled at intervals of 0.01 (from -10 to 10) and 0.1 (from -100 to 100), respectively. The resulting waveforms are organized into a 3×2 subplot grid for comparative analysis.

1.1.1 MATLAB Code

```
t = -10:0.01:10;
omega = -100:0.1:100;
alpha = [0.1, 0.5, 1.0];
x = @(t, a) (abs(t) < a);
ft = @(omega, a) 2*a*sinc((omega * a) / pi);

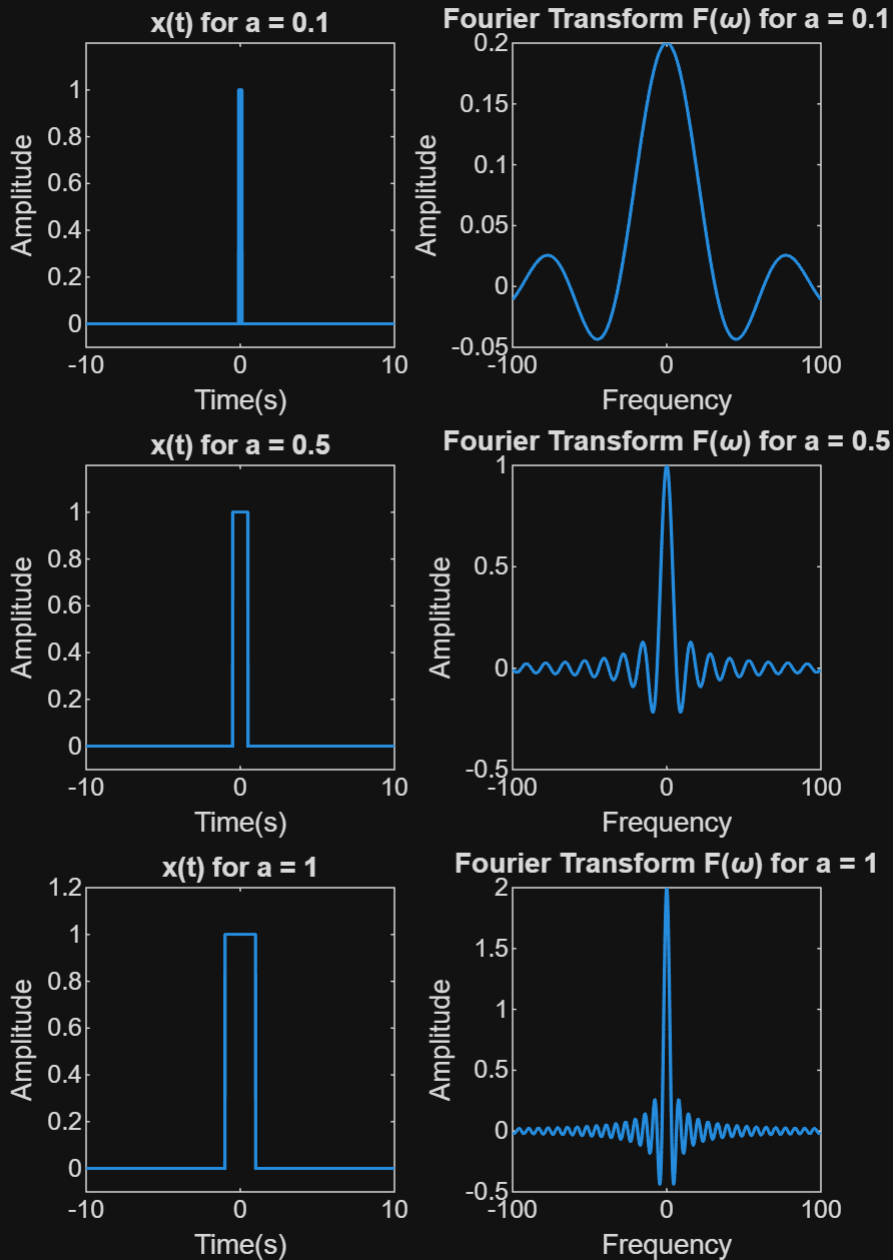
%% In the Math Problem section, I evaluate that the Fourier Transform of
%% x(t) is just (2*a*sin(omega * a))/(omega * a). However, according to the
%% officail documentation("https://jp.mathworks.com/help/signal/ref/
sinc.html#mw_e3c5bb46-ffb8-4805-9512-17c003c986c7"),
%% the final angle in sinc() is equal to the product of pi and the radius.
%% So that I divided the argument by pi.

cnt = 1;

figure('Position', [100, 100, 600, 900]);
for idx=alpha
    subplot(3, 2, cnt); % Select the first subplot
    p1 = plot(t, x(t,idx), 'LineWidth', 1.2); % Plot the Fourier transform
    for the given 'a'
        xlabel("Time(s)");
        ylabel("Amplitude");
        ylim([-0.1, 1.2])
        title(['x(t) for a = ', num2str(idx)]);
        cnt = cnt + 1; % Increment counter for the next subplot

    subplot(3, 2, cnt); % Select the first subplot
    p2 = plot(omega, ft(omega, idx), 'LineWidth', 1.2); % Plot the Fourier
    transform for the given 'a'
        xlabel("Frequency");
        ylabel("Amplitude");
        title(['Fourier Transform F(\omega) for a = ', num2str(idx)]);
        cnt = cnt + 1; % Increment counter for the next subplot
end
sgtitle('Fourier Transform Analysis');
```

Fourier Transform Analysis



1.1.2 Discussion

The Fourier Transform charts shown above indicate several key features.

First of all, as the variable a increases, the width of the rectangular pulse $x(t)$ in the time domain becomes wider. Conversely, the width of the main lobe in the Fourier Transform $X(\omega)$ becomes narrower, and its oscillations occur more rapidly. Furthermore, the peak value of $X(\omega)$ at $\omega = 0$ increases proportionally, which is exactly equal to twice the variable a ($2a$). From this result, it is evident that there is an inverse relationship between the time and frequency domains in terms of signal width.

Theoretically, if we expand this to the condition where a approaches infinity ($a \rightarrow +\infty$), the time-domain signal $x(t)$ would become a constant DC signal of 1. Correspondingly, the peak of $X(\omega)$ at $\omega = 0$ would approach infinity, and its values at $\omega \neq 0$ would converge to 0. This perfectly illustrates that $X(\omega)$ approaches an impulse function.

1.2 Math Problem

Below is the snapshot of my answer in fill-blank math problem.

The image shows a handwritten derivation on lined paper. The first line shows the integral for the Fourier transform:
$$X(\omega) = \int_{-a}^a x(t) e^{-j\omega t} dt = \int_{-a}^a e^{-j\omega t} dt$$
 The second line shows the evaluation of the integral:
$$= \left(-\frac{e^{-j\omega a}}{j\omega} + \frac{e^{j\omega a}}{j\omega} \right) = \frac{e^{j\omega a} - e^{-j\omega a}}{j\omega}$$
 The third line shows the final result using the sine function identity:
$$= \frac{2 \sin(\omega a)}{\omega}$$